

Technical Document #256: Cloud Computing - Comprehensive Overview

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Content delivery networks (CDNs) distribute content globally to reduce latency. CloudFront and Cloudflare are popular CDN providers.

Kubernetes orchestrates containerized applications across clusters of machines. It handles deployment, scaling, and management of containerized workloads.

Microservices architecture structures applications as a collection of loosely coupled services. Each service runs in its own process and communicates via APIs.

Serverless computing allows developers to run code without managing servers. AWS Lambda and Azure Functions automatically scale based on demand.

Multi-cloud strategies use services from multiple cloud providers. They reduce vendor lock-in and optimize for cost and performance.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Bioinformatics applies computational methods to analyze biological data. It has accelerated drug discovery and our understanding of genetic diseases.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Quantum computing promises to solve problems that are intractable for classical computers. It has potential applications in cryptography, drug discovery, and optimization.

Key Takeaways:

- Platform as a Service (PaaS) provides a platform for developing and deploying applications.
- Microservices architecture structures applications as a collection of loosely coupled services.
- Content delivery networks (CDNs) distribute content globally to reduce latency.